

Project 2, MA346, Spring 2026

Summary

In this project, you will do a data exercise that requires you to merge two datasets and do a statistical analysis on the resulting, merged data. You can choose which two datasets to merge and what question you would like to use them to answer. This document lists requirements that you must fulfill, so that your chosen project includes enough of our course content to show off what you've learned. Within those limits, you are free to choose the topic, the dataset, and the methods.

Due date

The due date for this project is April 29, 2026, by 11:59pm. This is intentionally *before final exams*, so that you have fewer things to do all at once and I have more time to grade large projects. Our final exam block will be used for office hours supporting a final homework assignment and any make-up quizzes students need to take.

Partnering

You should work with one partner on this assignment, that is, in teams of two students. If you are having trouble finding a partner, I can help you find someone else who is also looking. If there end up being an odd number of students needing partners, I can authorize one team of three.

Specific Requirements

Your project must include each of the following accomplishments.

- **Choose a question you want to answer with your project.**

- The report you will produce (described in greater detail below) should state at its outset your chosen goal and how you achieved it. It should show your results using whatever means make sense for your question (a table, a graph, the results of a statistical test, etc.).

💡 Example questions students have approached in the past

- Do Boston cops give out more traffic tickets when the Red Sox lose?
- What are the most significant variables in predicting survival for people involved in motor vehicle accidents?
- Is there a relationship between unemployment rate and the frequency of violent crimes?
- How do words used to describe wines relate to the quality scores that sommeliers assign to those wines?
- How did the most-used words in Yelp reviews change, when comparing reviews submitted during the height of the COVID pandemic to reviews submitted after it?
- What factors seem to have the greatest impact on whether a high school student will complete an application for government financial aid?
- How accurately can we predict the market value of players in European club football (soccer)?
- Do more automatable occupations tend to command lower wages, and how do occupational skills influence that relationship?
- Does the release of new original shows impact Netflix stock price?
- Which key statistics for each NFL player position are most closely correlated with the value of a player's contract on their first season on a team?

- **Find and prepare your own datasets.**

- I will not assign a topic or data for this project. You choose your own topic and find relevant datasets. There are many good data sources online, as we have discussed in the past. Be sure that you know and approve of the origins of your data, rather than just accepting a dataset someone you've never met posted to a platform in which anybody can post anything.
- Data cleaning is required only if the datasets you find happen to need it. If you're lucky enough to find data in the perfect format (which is rare) that's lucky for you. Data cleaning is not a requirement of the project, but most projects end up needing it.

 No fake data

Fake or synthetic data is not permitted in Project 2. Here are the reasons why:

- It's almost always too clean, which lets you avoid showing many of the skills you learned in this class.
- You can't draw realistic conclusions from fake data, because fake data has no value in the real world. The purpose of data analytics (and of a good Bentley education!) is to help you contribute value to the world.

If you're not sure how to tell whether a data set you found online is synthetic, refer back to Chapter 12 of the course notes.

- **Demonstrate that you know how to merge/join data.**

- Your project must include at least two datasets *from two different sources*, and you must merge or join them in a logical and useful way. (The reason this requires getting data from two different sources is because merges of data created by the same people are often trivially easy, which doesn't allow you to show the full breadth of the skills we learned about merging.)
- Examples:
 - * In sports, merge player performance data in each year with other information from the same year (e.g., player salary, player's team success, etc.)
 - * In business, merge an organization's financial data with less quantitative information about the organization (e.g., sector, policies, scandals, new products, etc.)
 - * With data about countries, merge health-related measures with economic-related measures

- **Include at least one statistical method.**

- Ensure that your project uses at least one statistical method as part of your work. Ensure that the statistical method you choose to use is relevant and helpful for achieving the goal of your project that you state at the outset.
- You've had at least ST113 before MA346, and possibly other statistics courses as well. All ST113 content is accessible [in Python at this webpage](#), giving you various statistical tools at your disposal, including hypothesis testing, linear modeling, and more.
- In our course, we also learned some related skills, such as using exploratory visualizations to determine relationships and correlations among variables of interest, testing variables for normality visually, and performing simulations.
- Example methods you might use:
 - * Students doing the NFL project described earlier might compare variables with a heatmap of correlation coefficients.

* Students doing the Yelp reviews project described earlier might use a hypothesis test to compare two population proportions.

- **Include a report** suitable for reading by a potential employer.
 - Make the report something you're so proud of that you could post it on your LinkedIn and expect potential employers to visit it and see what you've learned.
 - Keep in mind that the target audience is a potential future employer. Don't write jargon understandable only by me (or a classmate who's studied what you have). Speak to an audience who is intelligent and interested in data projects, but probably doesn't know Python.
 - This report can be in any suitable format (Word, PDF, perhaps a slide deck if it fits the project and is done well). Although I have seen students attempt to put the whole report inside a Deepnote notebook, this rarely fits the target audience, which is supposed to be non-Pythonistas.
- **Publish your work online**, including your code, data, and report.
 - This is probably easiest to do by publishing a Deepnote project and placing all the relevant files into it so that I can download them from there. But you can publish it in any way that makes sense to you and gives a reader access to all the content. For example, you could choose to use a GitHub repo, a Google Colab project, or any cloud provider you like.
 - In addition to including your report in your published project, please also describe in your notebooks any data cleaning, visualization, and analysis code that you wrote, so that I can verify that your code does what is intended.
 - **Submit your work by placing the URL to this published project into Brightspace.**

 Access permissions are a common problem

Be sure to set the project's permissions correctly so that I can view it! *Many* students forget this. It's very easy to check; have your roommate or another friend (other than your teammate for this project) try to view the project. If they can see it, so can I (probably).

Where to find data

If you have a topic you're considering: Start with a web search like “free golf datasets” or “free nutrition datasets” or whatever your topic is. Don't take just the first result; persist until you find data that is comprehensive enough, clean enough, and reliable enough for your needs. Remember that synthetic data is not allowed.

If you're not sure what to merge that data with, look at the variables that exist in the first dataset you've found, and ask which of them might make sense for merging with another dataset.

If you don't have a topic you're considering: Try browsing some of the free online data repositories we've discussed earlier in this course. A web search or a chat with your instructor can help you find large repositories of huge numbers of free datasets on tons of topics.

How you will be graded

Use the following grading criteria as a checklist to be sure you've done everything expected of you for this project.

Basics:

- Submitted a link to your published project by the deadline. 15%
- Your published project includes a report in any of the acceptable formats. 5%
- Your published project includes any Python code you wrote as part of the work. 5%

Writing:

- Your report begins by stating your goals and summarizing your results, then ends with any other necessary details about the work you did. 5%
- Your report explains the origins of your data, why it fits your goals, and what you had to do to get it into a suitable state. 10%
- Any code notebook you submitted follows the before-and-after principle for explaining itself (see course notes Chapter 2 for a review of this concept). 5%
- Your report was suitable for the target audience you've been assigned 10%

(Recall that the target audience is a future employer browsing your LinkedIn profile or resume: They're intelligent and interested in data projects but don't know Python.)

Math and Python:

- You found two datasets that it made sense to merge, and merged them in a way that was logical and was also helpful towards your goals. 15%
- Your data cleaning and/or EDA steps were done correctly and logically. 15%
- Your work included some type of statistical analysis (test, modeling, etc.) and that work was done correctly. 10%